To Investigate the Design and Manufacture of A 5” Gauge Locomotive Chassis and Evaluate its Performance

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Abstract

This project details the process of designing and manufacturing a 5” gauge locomotive; focusing on the creation of a Class 08 chassis, using research alone. The project was chosen because the design and manufacture of a model locomotive is a sophisticated process which requires specialist skills and knowledge, that can be acquired through analysing various resources. These resources ranged from images of the full-size locomotive to books that detailed the possible manufacturing processes which could be used to bring the formulated designs into reality. Through acting on new findings and following an iterative design process, I was able to produce a running chassis which recreates the design of a Class 08 to a degree of accuracy.

Introduction

Miniature locomotives are highly detailed, meticulously engineered scale representations of their full-size counterparts; which depending on their size, are powerful enough to pull people around a track. Many types can be recreated, from intricate steam engines that can take experienced engineers lifetimes to build, to simpler less complex diesels. Therefore, I chose to produce a diesel locomotive for my project; in particular the Class 08 due to its versatility, relative simplicity and portability compared to larger diesels. The Class 08 is a shunting locomotive, often found at stations and freight yards as some 966 engines were commissioned (“British Rail Class 08”, March 2023).

Figure 1: A Class 08 (Avonvalleyrailway.org, March 2016).

The commonality of the locomotive generated a range of resources such as images and technical data which permitted me to formulate drawings that I used to manufacture the locomotive in my home workshop. The generation of these drawings myself was the main aim of my project because replica Class 08s have been built before using commercial drawings (blackgates.co.uk, 2013). Consequently, I wished to investigate the performance of a locomotive that has been designed and manufactured based on my research alone, to highlight the feasibility of using original drawings in the field of model engineering. This approach allowed me to realise how the mechanical systems that drive locomotives operate, emphasising the engineering behind their design and production which is an area of personal interest.

Thus, my initial objectives were to utilise various research methods to manufacture and create drawings for a suitable locomotive that could be assessed based on its aesthetical and mechanical qualities. Hence, this would enable me to determine whether an accurate model could be reproduced using research. Achieving these aims developed my technical drawing, problem solving and engineering skills. However, I realised the true extent of my project once I had produced some of the drawings, so I decided to focus on the creation of a working chassis that would pull at least 2 people around a track.

Methodology

Most of my research followed a linear path as shown by my Gantt chart. The first research that I completed was based on the locomotive design, ready for creating the drawings. This research constituted a large portion of my project because the success of other sections was reliant upon accurate technical drawings. Once I had produced the drawings, I researched possible manufacturing processes ready for machining the components. However, because building a locomotive is an iterative process, I often had to deviate from this linear approach and revise my drawings based on further research.

Figure 2: A Class 08 model (Stationroadsteam.com, 2010).

Researching The Drawings

To produce the drawings, detailed insight was required into the correct design, size, manufacture, and location of certain Class 08 features. This was partly achieved through analysing images of the full-size locomotive and 5” gauge models below.

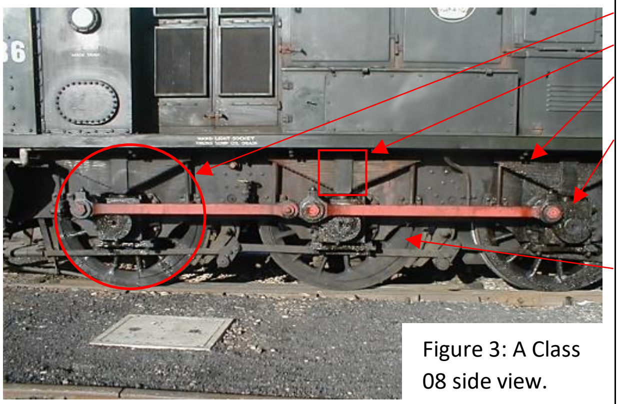

Figure 3: A Class 08 side view.

The frame width can be worked out using the diameter of a wheel, which is slightly less than the height of two axle boxes.
Suspension is provided by leaf springs which are integral to the aesthetics; but replication may be difficult, so I will use simpler blocks.
The wheels are coupled with pear-shaped cranks that drive a front coupling rod, which connects to the shorter trailing rod via a half-lap joint. Further inspection shows oil wells above the crankpins; this overall shape will be implemented into my drawings.
There are curved cut-outs between the axle boxes and leading up to the buffer bars. These radii will be added to the frames that I design because integration is easy and it will the improve aesthetical accuracy (Barry Leach, July 2006).

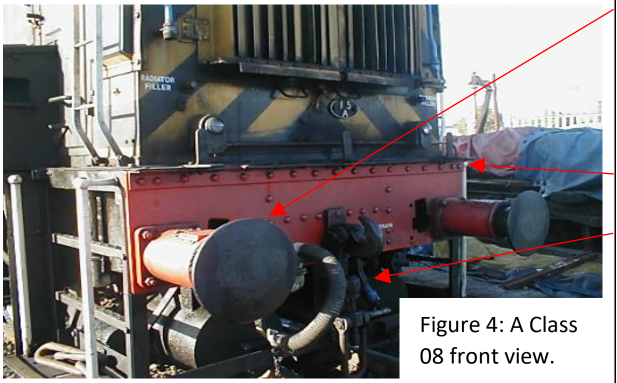

Figure 4: A Class 08 front view.

The buffers consist of a sprung round head and a stock with a square mounting plate. I synthesized this image with Figure 11 to determine external dimensions such as the 150mm front radius. However, the placement of the buffers at the ends of the buffer bar may not be feasible with a standard buffer spacing dimension of 6”.
Round head rivets fix the components together, so throughout the chassis I will use round headed fasteners (Sandman, R, 1962, page 85).
Further detail is added with a central screw link coupling and vacuum hose. This can be recreated on the front buffer bar, however on the rear buffer bar a clevis and bar coupling will be used to reduce the risk of uncoupling (see page 36), (Barry Leach, July 2006).

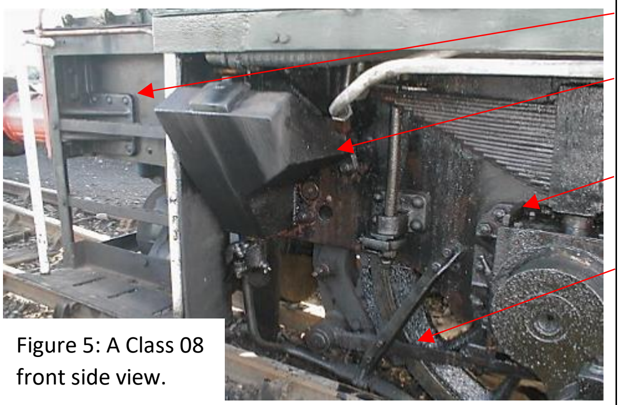

Figure 5: A Class 08 front side view.

There is a buffer bar support bracket which prevents the frames from twisting. This should be recreated to improve strength and accuracy.
The location and shape of the front sandbox can be determined. Though, the shape will be simplified in my drawings to make machining easier.
The axle box guides (horn blocks) are cast and riveted on the outside of the frames, but I will use the simpler versions detailed in Figure.9 because machining castings is tricky.
The brakes consist of various linkages which could be implemented into my drawings. However, this would increase the complexity.
The access steps are an integral feature, though to save time I will not replicate them (Barry Leach, July 2006).

Fig 6

Fig 7

Figure 6 shows the whole locomotive which helped me to understand general arrangement of features along the chassis. Figure 7 details the slightly different shape of the rear sandbox and frame cut-outs, as compared to the front. This will enable me to create correct location-specific drawings (Barry Leach, July 2006).

Figure 8: A 5" Gauge model

The front buffer bar is like the full-size locomotive, with buffers and a coupling hook. However, the rear buffer bar must use a clevis coupling for safety.
The suspension springs are held captive with simple blocks that would be easier to manufacture than intricate leaf springs; I will replicate this.
Located halfway between the axle boxes are screws that retain reinforcing frame stretchers. These will be implemented for strength.
Using this model and others alongside the full-size Class 08 enables me to work out scaled dimensions and spacing, for example the distance to the front axle from the leading edge of the frames.
The wheels are spoked like the real loco, but replicating this would require complex CNC machining. So, I will use solid wheels like the model below (Maxitrak.com, 2020).

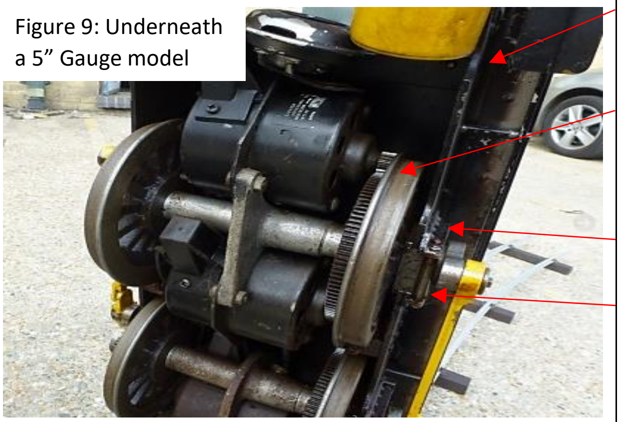

Figure 9: Underneath a 5" Gauge model

The frames are 4mm thick for strength and incorporate the curved cut-outs seen on the real locomotive. These features will be designed primarily using the images of the real loco to improve precision.
Each axle is driven by two geared motors that fit within the constraints of the chassis. These motors are mounted to the axles to prevent the gears from meshing incorrectly due to the vertical movement from the suspension. This system is complex so other options will be considered.
The horn blocks are created simply, using two pieces of steel angle which are riveted to the frames to provide lateral axle support.
The axle boxes are retained in the slots with a small round pin that is screwed into the axle box guides; I will use this straightforward design.
The coupling rods are retained with small screws onto the crankpins (Maxitrak.com, 2020).

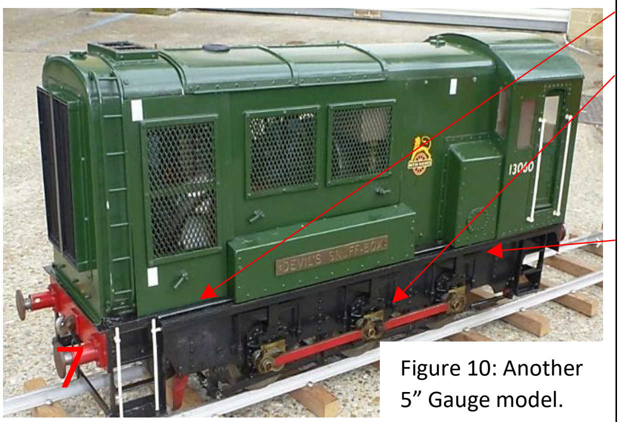

Figure 10: Another 5" Gauge model.

The footplates are around 6mm thick, like the real locomotive but these would have to be manufactured from aluminum to reduce weight.
The axle boxes are made from solid bronze with no separate bearings, this may cause issues with maintenance and the high cost of bronze. So, I will make mine from steel with separate bronze bearings.
This model uses a petrol – gearbox power system, which proves concept viability. However, utilisation is trickier than battery-electric.
Details such as leaf springs and steps enable close replication of the real class 08 chassis even though there are no sandboxes. The details that I integrate will be based on their complexity versus aesthetic gain.
Overall, these main design points were invaluable for replicating component shapes, hence proving the adequacy of these images (Maxitrak.com, 2019).

These images were gathered from Cheltsme.org and Maxitrack.com (which is a selling page), but the reliability of these images was not a concern because there was little risk of them containing bias or incorrect information.

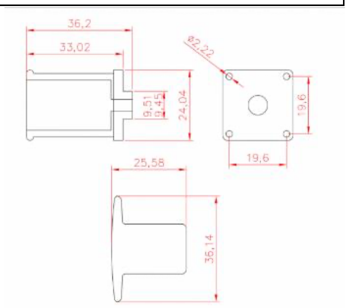

Figure 11: Dimensions of 5" gauge buffers (17d-ltd.co.uk, 2023).

Yet, these images only provided clues on power systems and the basic shape of components; therefore, I carried out further research to investigate possible drive systems, standards, materials, and mechanical parts such as bearings. This was achieved using information gathered from various websites which were assessed for validity using the Raven and CRAAP models in my appendix. These assessments highlighted the risk of bias in this information, because many of the websites had vested interest in selling their products. The main findings from this research are outlined below.

Standards

To produce components that operate correctly, certain standards must be integrated into the drawings. The most crucial standard relates to the design of the wheel tread. If the tread profile consisted of a simple flat section and perpendicular flange, the flange would tend to slice the top of the rail while traversing a curve. This would result in increased wear or derailment; this is called side cutting.

To mitigate this, a more complex profile is utilized as shown by Figure 12. In addition, the distance between the wheels is stated as 4 11/16”; this leaves some clearance between the rail and flange while traversing curves. Which, when combined with the correct tread profile reduces rolling resistance and side cutting, improving the overall running of the locomotive.

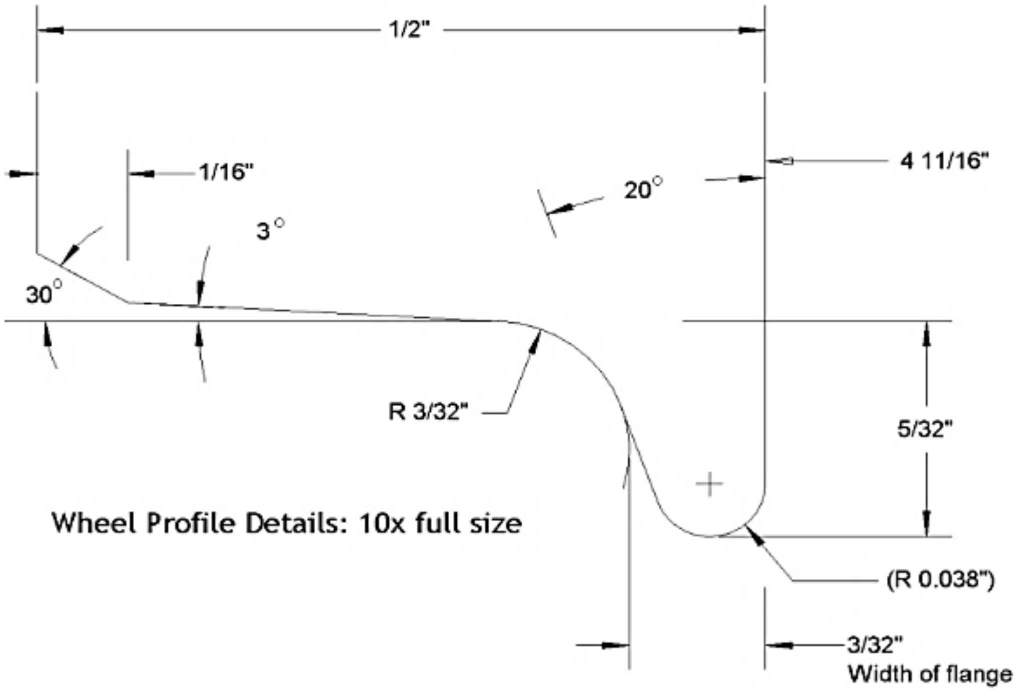

Figure 12: The standard wheel tread profile and wheel back-to-back distance (Gl5.org, March 2023).

Another standard that I employed relates to the buffer and coupling location, which ensures that they line up with other locomotives. As shown by Figure 13, the buffers should be 3 5/8” above the rail head and spaced 6” apart.

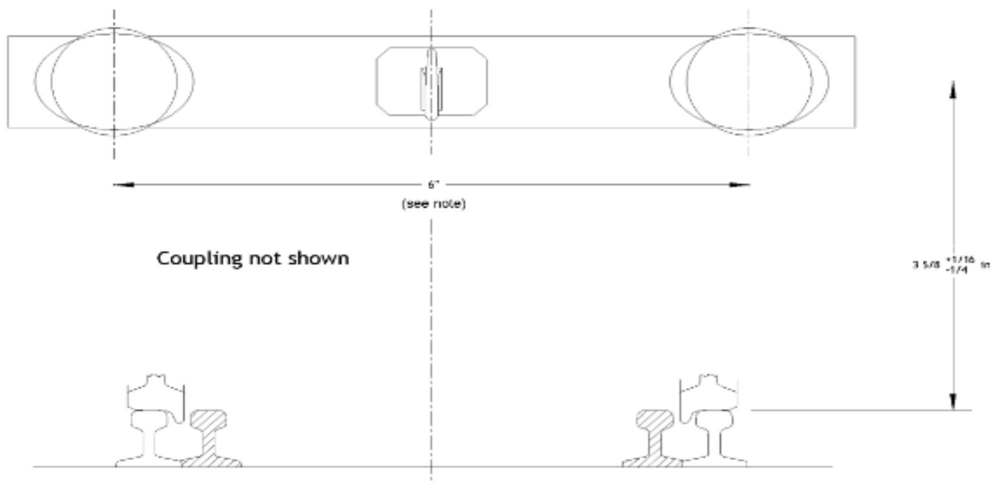

Figure 13: The 5” gauge standard Buffer height and spacing (Gl5.org, March 2023).

These standards were attained from the Ground Level 5 Association's website, as a reputable source of tested and peer reviewed data gathered by experienced engineers. Though, these standards lacked certain details such as clearances, so I amalgamated them into my drawings with an article from my local model engineering club's website to ensure that my locomotive would run smoothly around their track (Monk, 2004).

Mechanical Systems

As seen in the photos above, the axle boxes require suspension using compression springs. These springs had to be chosen before the drawings were created, based on their dimensions and rate (which relates to their stiffness and is measured in Newtons per mm). The rate was worked out by taking the resting weight of the locomotive, minus the wheelset weight and dividing this by 6 (6 axle boxes). This is the resting force per axle box, which must compress the spring by 3mm; so, the axle box sits halfway up the slot (6mm) to compensate for track irregularities. Therefore, to find the rate, the resting force per axle box should be divided by 3. Using this method from parksiderailways.co.uk (2015) with a resting weight of 500N results in a rate of 27N/mm, which allowed me to choose suitable springs from springmasters.com (2023).

Figure 14: A ball bearing, used for the layshafts (Simplybearings.co.uk, 2023a).

A bearing is required wherever there is a spinning axle in my drawings, the type chosen for each situation depends on the forces applied. In my designs I could use plain or ball bearings for radial loads and thrust bearings for axial loads. Ball bearings have the greatest reliability, however plain bearings can also counteract axial loads (if combined with a flange) and are simple enough to design/ manufacture in house out of SAE-660 leaded bronze to specific requirements. So, I will use these for most applications in my chassis except for the lay shafts where ball bearings will be used because access for oiling is limited (Andreas Velling, August 2020), (Leedsbronze.co.uk, 2021).

Figure 15: A flanged bearing (Simplybearings.co.uk, 2023b).

The use of bearings is crucial in the transmission; which transfers motion from the motors to the axles via either gear, chain, or belt/pulley systems. These systems are applicable because they can alter the rotational output speed and torque. The most suitable transmission for my chassis is either a chain or pulley system because the sprocket placements are not as critical compared to a gear system, which requires perfectly meshing teeth for efficient energy transfer. The flexibility of chain and pulley systems enables frame mounted motors, which reduces complexity while enabling further speed reduction than axle-mounted motors. However, belt and pulley systems rely on friction which can be overcome when under high loads. Therefore, I will use a chain drive system from simplybearings.co.uk(2023c) with the correct speed reduction ratio in my chassis (Yohan, May 2018).

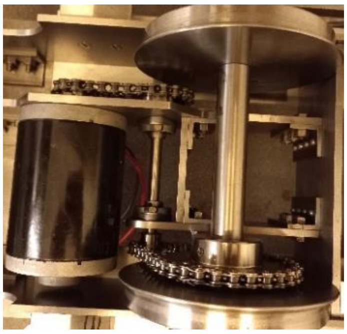

Figure 16: A chain drive system with a lay shaft.

This transmission system must have brakes for safety. Possible types vary from electronic regenerative to disc brakes (Figure 17). Disk brakes are effective; however, they require space and an activation system which adds intricacy. Therefore, I will use the built-in regenerative braking provided by the speed controller, which works by using the motors as generators to recharge the batteries when the demand speed is lowered beyond the actual speed, this creates a resistive force that slows the chassis down (4qd.co.uk, 2023a).

Figure 17: Disk brakes for 5" gauge locomotives (17d-ltd.co.uk, 2023).

Power Systems

Some source of energy is required to drive the transmission system. Petrol or battery electric power systems would be suitable, though the system must provide control over the output RPM, because the axles will be driven via a fixed speed reduction ratio. An inherent issue with petrol is that at low RPM the output power reduces, therefore the output speed should be kept constant, within the most efficient range. To achieve this the engine would connect to a gearbox or generator/motor system to regulate the output speed. A possible petrol engine could be the small 4 stroke Honda GX25 (Figure 18) which would fit inside the bodywork. However, modifying the output RPM is complex, so a battery system is more applicable. Battery systems provide fine control over the output speed via an electronic speed controller, this varies the battery potential difference though pulse width modification; which regulates the motor shaft speed. To determine the best solution, I completed some testing that involved connecting the engine and a possible generator together. I ran the engine at different RPMs, measuring the output speed of the generator-powered motor (Figure 19). This highlighted the difficulty of routing exhaust gases, failing couplings and a very poor energy transfer efficiency. Therefore, I chose the simpler battery electric system (Maxitrack.com, April 2008).

Figure 18: Honda GX25 Engine (engines.honda.com,2023).

The following research was used to determine the components required for a battery-electric system. Batteries were selected based on the average current drawn by the traction motors, as high currents can cause damage and reduce the overall battery capacity (as shown by Figure.20). To find this average current I worked out the rated motor current using current = (Power(150w) ÷ voltage (24)) x 2 = 12.5 Amperes), I multiplied by two because the motors are wired in parallel. Using this rated current the average current could be found by summing the 1-minute currents in Figure 21 (to get the average running current), and multiplying this by 0.7 which is a ratio of moving time/ total time. This results in an average current of 6.3 Amps drawn by the motors, which when used against the graph shows that a battery capacity of 40-amp hour would last 5 hours. So, I will wire two 12V 40 AH batteries in series to gain the required 24v.

Figure 19: Testing the petrol-electric power system.

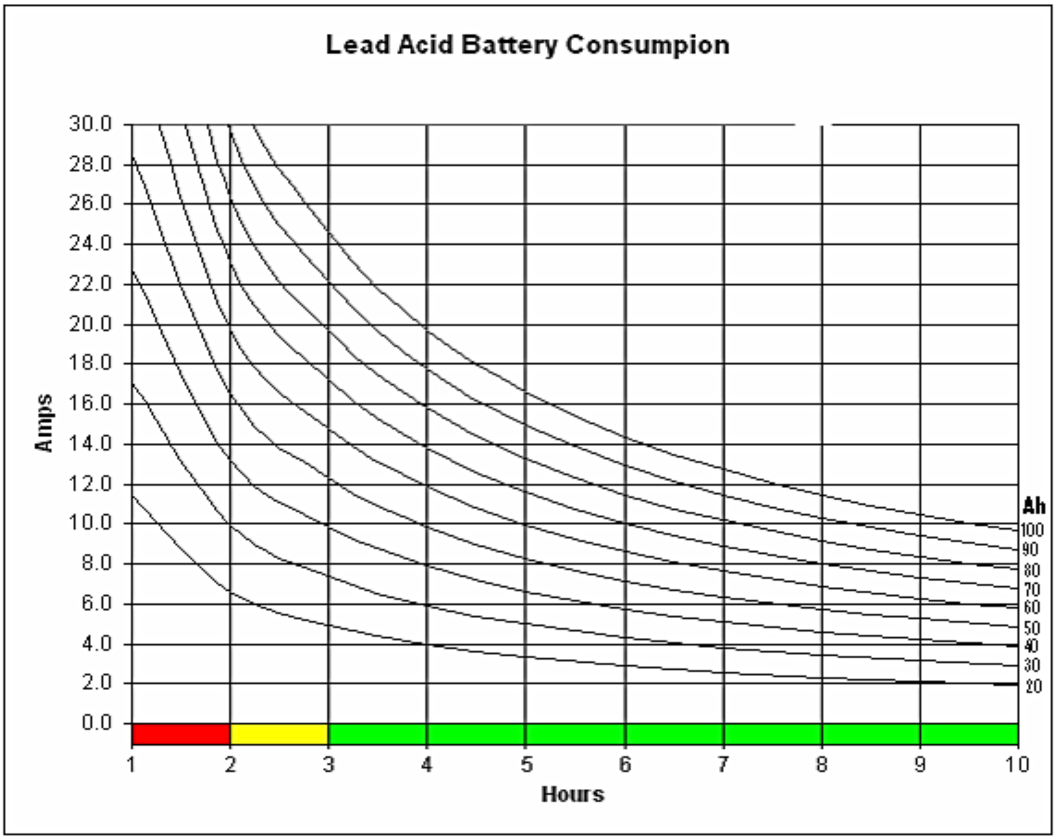

Figure 20: A graph showing the relationship of battery capacity to the average current drawn and running time (Maxitrack.com, April 2008).

Running condition	Percentage of time on the track	Current drawn at running condition	1-minute current
Pulling away	10%	24 amps (2xrated)	(24 x 0.1) = 2.4
Up a gradient	30%	12 amps (rated)	(0.3 x 12) = 3.6
Down a gradient	30%	2 amps	(0.3 x 2) = 0.6
Flat	30%	8 amps	(0.3 x 8) = 2.4

Figure 21: A table used to work out the average current.

To choose the motor electronic speed controller, circuit breakers and wire size the maximum current is used. This is the current drawn by the motors when accelerating from rest (maximum stall condition). It can be up to 3x the rated current; which in my case is 38 Amps (4qd.co.uk, 2023b). The two possible manufacturers of controllers are 4QD and Mtroniks, who sell the DNO 10 and Dci75-24 respectively; both come with a handheld controller and handle more than 38 amps. As shown by the table, the specifications are very similar. However, the DNO 10 is cheaper and has equal quality, So I will use the DNO 10 controller.

Controller	Price	Max Current (Amps)	24v Operation	Regenerative Breaking	Motor type	Reversing
4QD DNO 10	£160	100	Yes	Yes	Brushed	Yes
Mitroniks Dci75	£250	75	Yes	Yes	Brushed	Yes

Below, Figure 22 details how I will wire up the DN0-10 controller and handset. These wires will have to be carefully routed through the chassis. The main revelation from the diagram and supporting article is the importance of fault protection, provided by either in-line fuses or breakers. These would be placed in series with the motors and batteries to limit the risks posed by excessive current. The sources of this information were mtroniks.net (2023), 4QD.co.uk(2023c) and Baker (2022). While there was a risk of vested interest, the reliability of the data was not an issue due to the experience of the authors and strict editing of the model engineer magazine.

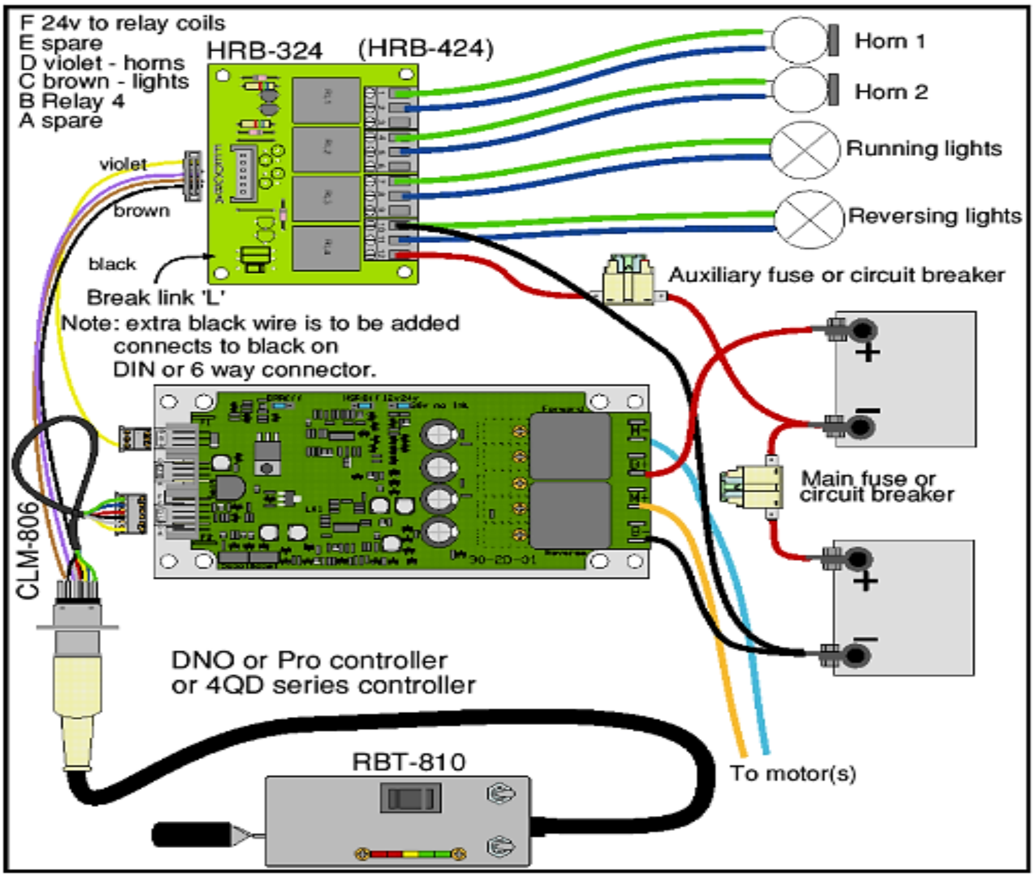

Figure 22: The wiring diagram for the DN010 Controller (4qd.co.uk, 2021).

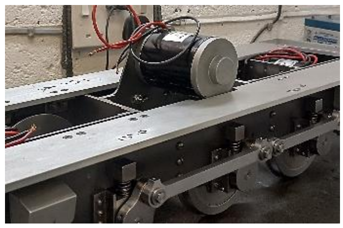

Figure 23: The incorrect third motor.

(2650 ÷ 6.6) = 401 Axle RPM
401 × 60 min = 24060 RPH

RPH × wheel circumference = the Meters traversed per hour (24060 × 0.11 × π) = 8310m

8310 ÷ 1609 (M in a mile) =

5.16 MPH

Figure 24: Working out the maximum speed.

The axles could be driven by brushed or brushless motors, in the correct dimensions. While brushless motors have greater reliability and efficiency than brushed, they are more expensive and require specialized electronics. Therefore, I will utilise brushed motors because they are easier to implement (powerelectric.com, 2020).

Figure 25: A Parvalux motor (parksiderailways.co.uk, 2020).

Figure 26 shows the chosen motors. These are cheaper versions of the commonly used 150w Parvalux motors (Wyatt,2015). While dependability is reduced, their availability and lower cost makes replacement a simple task. These motors have a high efficiency of 80%, so I could expect a power output of 240w from 2 of these 150w motors wired in parallel, which is sufficient for this size of locomotive (buggies.builtforfun.co.uk/,2011). However, all small motors have high spindle RPMs, so a large speed reduction must be employed. The prevalence of this was realized when I consulted an experienced engineer on the performance of my power system. They advised that the motor output speed should be reduced further to enable the use of an efficient 24v system. This information highlighted issues with my internet research, however it could have contained bias, so I critically assessed other solutions to this problem using model engineer/Facebook forums. This resulted in the removal of the third motor and direct chain drives (Figure.23) which had a ratio of 3.3-1 (9 to a 30-tooth sprocket). This enabled the integration of lay shafts and a 6.6-1 reduction ratio using 9 to 18 then 9 to 30 tooth sprockets, which should result in a maximum speed of 5.2 mph (Figure 24), improving the performance.

Figure 26: The cheaper 150w 24v motor (buggies.builtforfun.co.uk/, 2011).

Generating The Drawings

This research was combined with the technical drawing and problem-solving skills that I developed to produce designs for every chassis component. However, my research which mainly consists of images, contained little information on dimensional scaling. So, I utilised a smaller 3/8" gauge model and full-size locomotive data to gather dimensions such as wheel diameter (1.370m) which could then be scaled to 5" gauge (0.110m) ("British Rail Class 08", March 2023).

I produced the drawings in first angle projection using the British Standard 8888 system to maintain legibility (roymech.co.uk, 2018). Creating drawings was difficult because I had to ensure that current and future components fit together, while incorporating the correct shapes, features, materials, and standards. Further complexity was added with the requirement for interference/clearance fits between certain moving parts like axle boxes; the implementation of which was critical for smooth running and longevity. For example, a clearance of ≈ 0.5mm was required between the wheelset and inner bearing flange to enable curve traversal. This information was carefully selected from conflicting opinions in model engineer forums, Gilligan (2016). Parts such as axles were simpler to design than the frames, coupling rods and buffers/coupling hook. These required further insight, using drawings like Figure 28. The drawings showed how the mechanical systems operate on other locomotives, which alongside my own intuition facilitated me to reverse engineer their designs into my chassis. For instance, to ensure that the cranks were accurately quartered, I created a design which utilised grub screws that locate onto perpendicular flats on the axles, using information from Evans (1968, page.42). However, some errors were made using this approach, such as the wrong coupling rod shape. These were rectified in new iterations. Some of the drawings are outlined below.

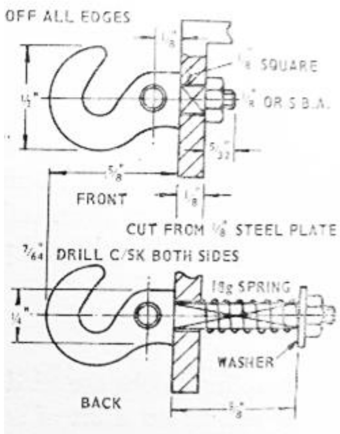

Figure 27: The sprung coupling hook design. (Evans,1968, page 129)

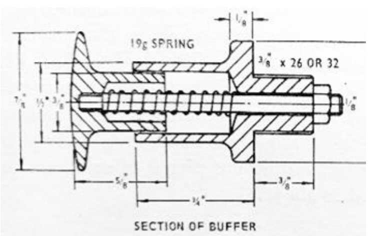

Figure 28: The internal buffer design. (Evans,1968, page.129)

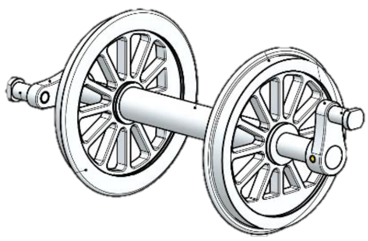

Figure 29: Quartered cranks of a Class 08. (Waal, 2017)

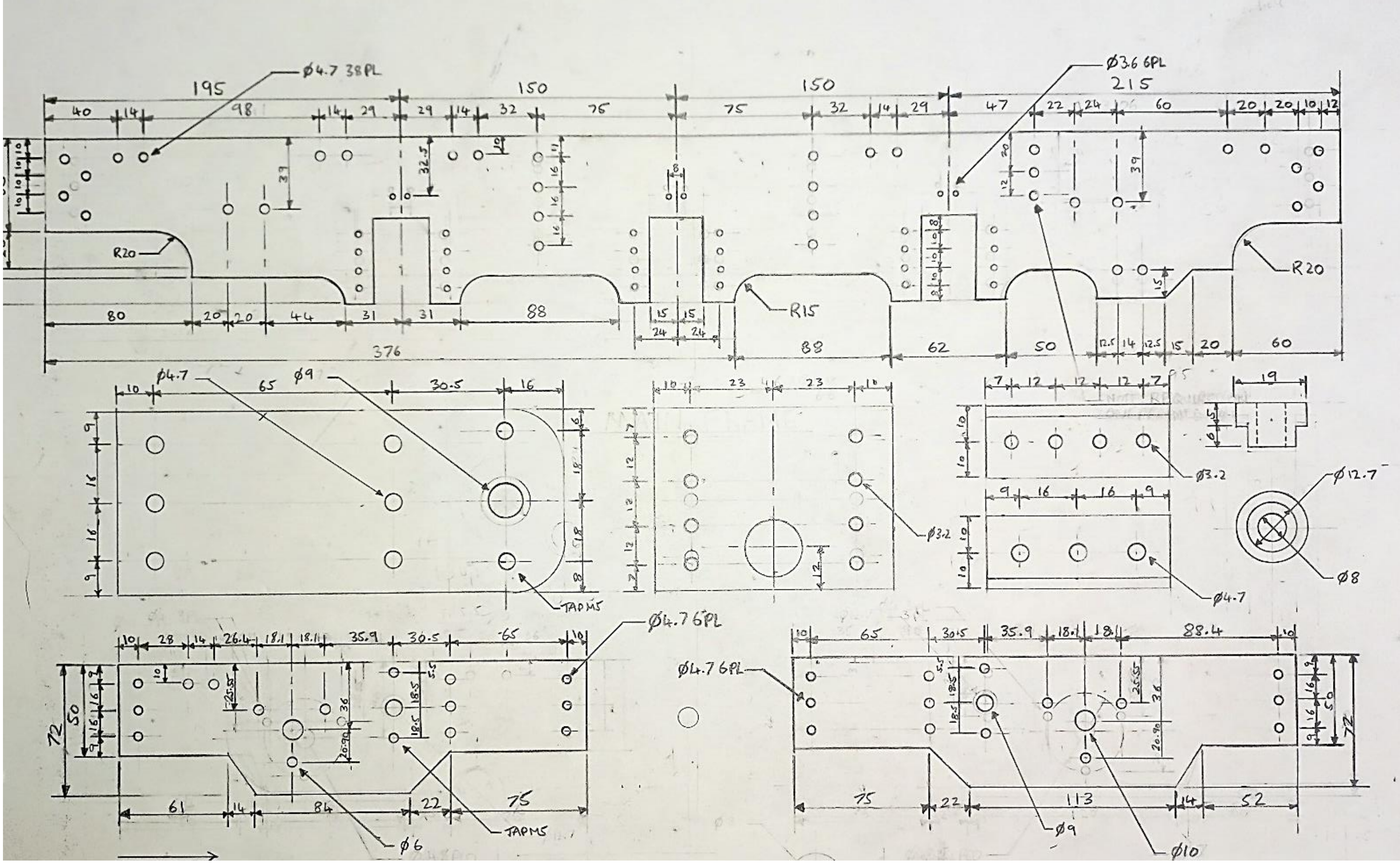

Figure 30: Drawings showing the main frame profile with the cutouts, replicated using the images; the frame width was worked out to satisfy the standard 3 5/8” buffer height. The transmission system frames are also shown, which hold the motors and roller bearings for the lay shafts in very specific locations to ensure the correct chain tension.

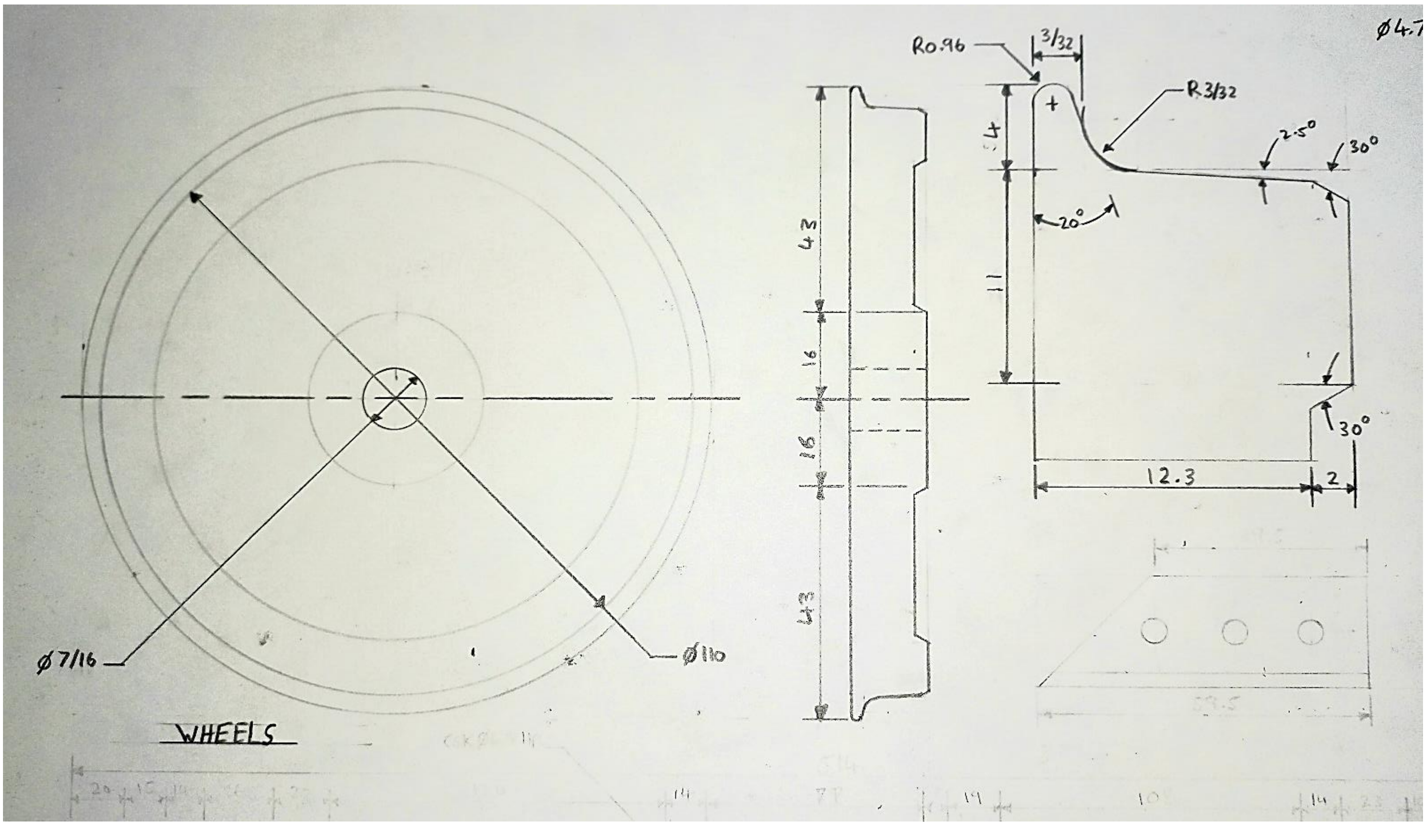

Figure 31: Drawings showing the wheel profile, which is identical to the standards, except the diameter is scaled from the full-size class 08 wheel.

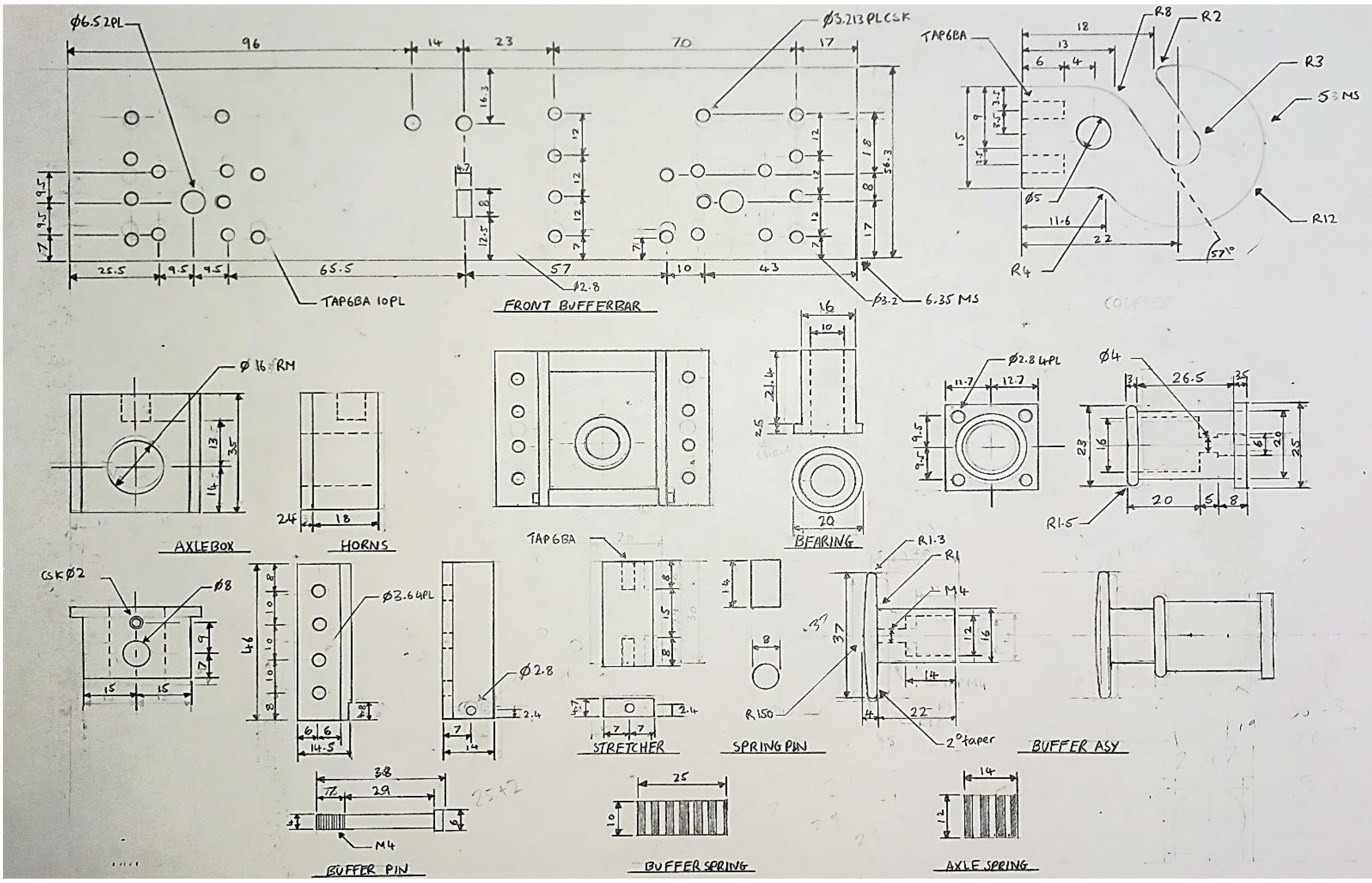

Figure 32: Drawings showing the front buffer bar, buffers, axle box and horn blocks. The design of the buffer internals closely follows the drawings in Figure 28. The axle box retainer is rectangular rather than round for stability.

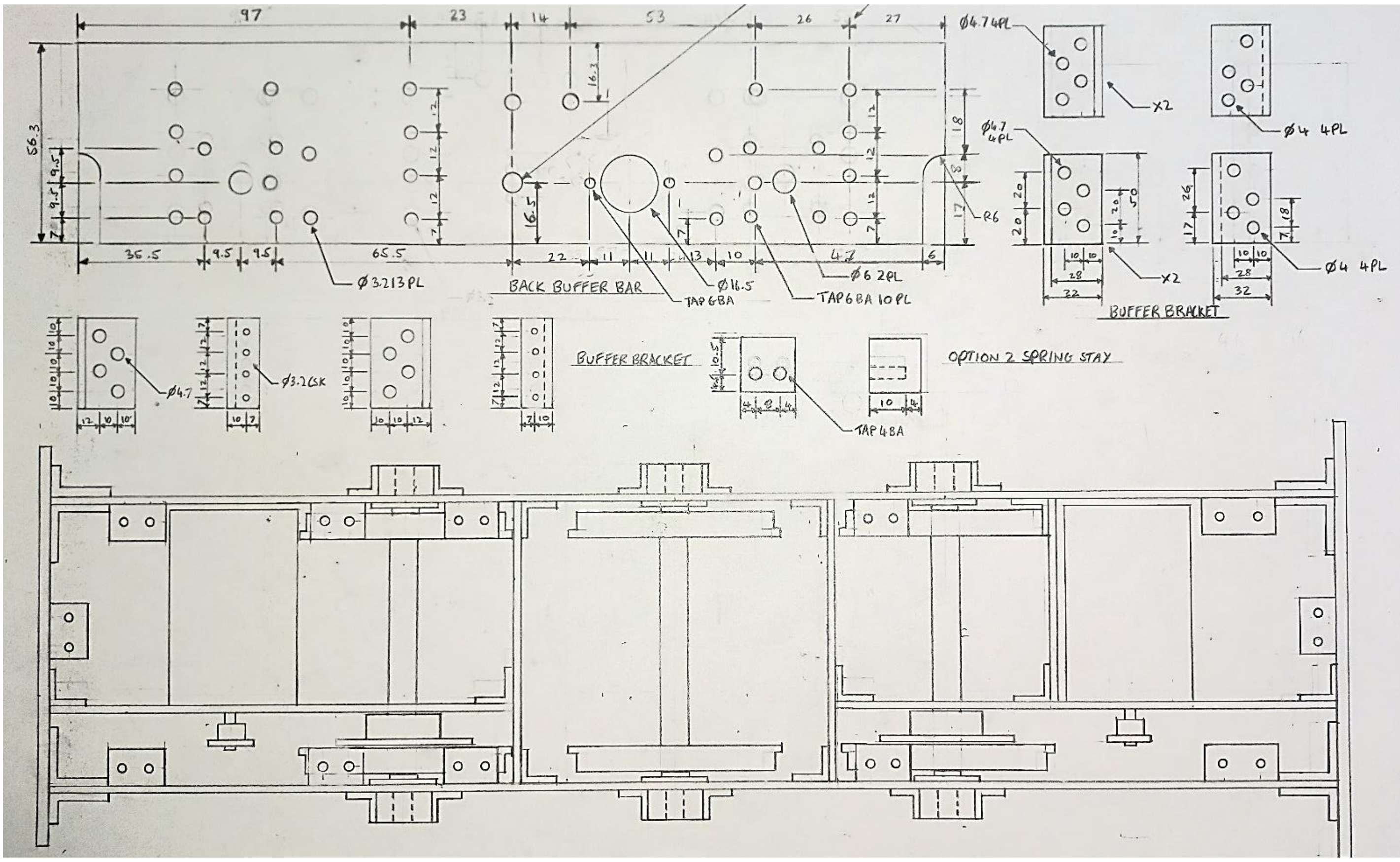

Figure 33: Drawings showing the rear buffer bar/ brackets, simple square suspension retainers and the initial general arrangement; all determined using the images of Class 08 models.

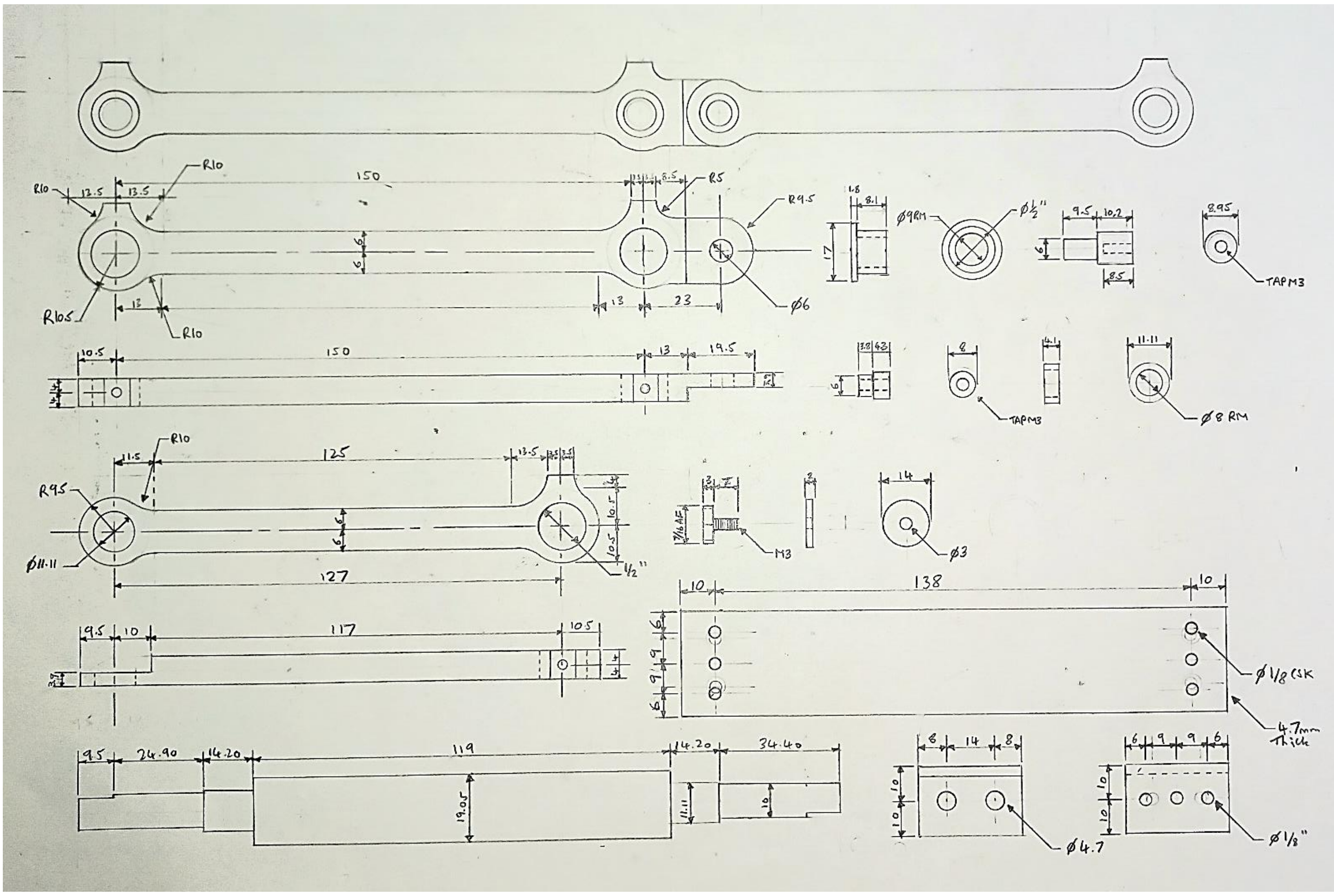

Figure 34: Drawings showing the axle, coupling rod bearings, frame stretcher and my second attempt at recreating the complex coupling rod profile.

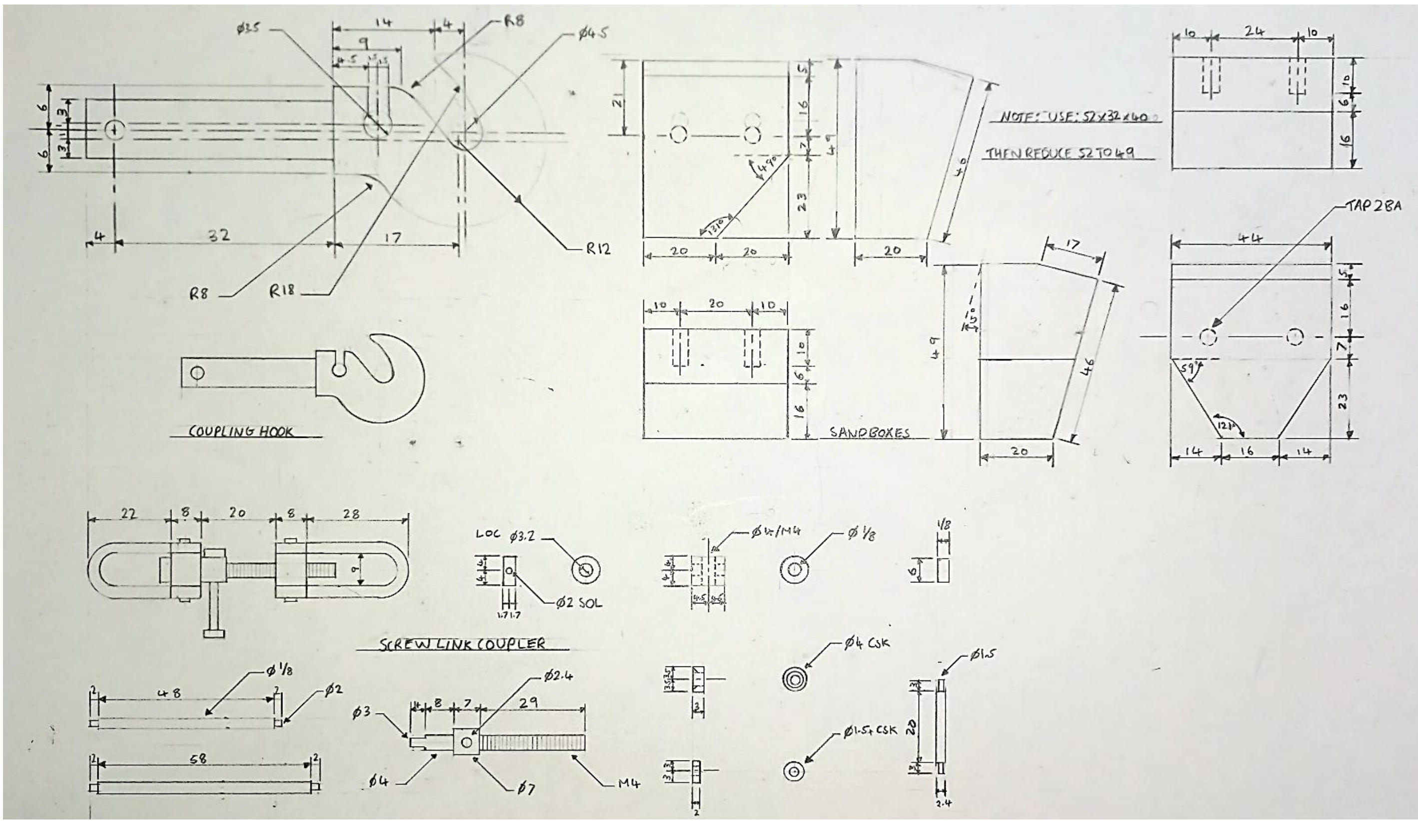

Figure 35: Drawings showing the front coupling, which follows the design in Figure 26 and the sandboxes; determined using the images and Figure 27.

Manufacturing

Once the drawings were finalised, I began the manufacturing research which utilised processes described in a book called "Simple Model Locomotive Building" (Evans 1968), these provided a guide for the machining of complex parts such as coupling rods. However, manufacturing processes are often specific to your expertise and available equipment so further ideas were gathered from model engineer forums/ magazines alongside YouTube videos. These resources were heavily influenced by individual views and opinions, which I took into consideration. However, as a machining process is based on the synthesis of many different approaches, each assessed based on appositeness, the prevalence of this issue was reduced.

Figure 36: A finished screw-link coupling, using Figure 35.

The material used for most components was steel because it machines well and comes in many types and grades. Cold rolled steel was employed for parts that require high geometrical accuracies like axle boxes, Hot rolled was used for the frames to minimise distortion from internal stress. The properties of the steel are determined by the grade, for example EN8 and EN1A have higher tensile strength than EN3B. So, EN3B would be more suitable for brackets than wheels or axles. Another possible grade is silver steel which is ground to tight tolerances, this makes it perfect for spindles like crankpins (Livesteammodels.co.uk, 2022), (Thurston, 2021) and (Sandman, R, 1962, page 66).

Other metals could be used, such as aluminium which comes in grades: 1050 and 6082. 1050 has a low strength making it useful for bending, 6082 on the other hand has a higher strength which makes it suitable for machined parts such as sandboxes (thyssenkrupp-materials.co.uk, 2018). Copper is a reddish-brown metal with high ductility and conductance. This makes it difficult to machine but great for wiring. Brass and bronze are alloys of copper and zinc or tin respectively; they machine well due to high tensile strength. Both Brass and bronze can be used in gears and bearings as a result of low friction coefficients; however, bronze is more widely used (Devi, 2021). All of these metals come in different profiles such as flat or round, the chosen shape will be based on the final shape of each component.

Figure 37: A wheel mounted using a mandrel.

Throughout the manufacturing process, I developed my machining/fitting skills, while following risk assessments for safety. Manufacturing was an exact science because tight tolerances of ± 0.025mm were required for machined parts. These were even smaller for bearings and spindles which had specific running fits of ± 0.005mm. Therefore, a few components had to be remade to satisfy these conditions.

The wheels were manufactured first, using free machining EN1A steel. They were complex to machine because each wheel demanded identical intricate tread profiles which were necessary for smooth running. To achieve this, each operation was completed on every wheel before the next commenced, using a mandrel to ensure concentricity (Figure 37). I chose to make the mandrel with a retaining screw to hold the wheels on rather than Loctite to enable quick changes. Once the billets were drilled, the tread profile was turned as per the drawings, with a round nose tool rather than a form tool to reduce vibrations and chatter. Once this shape was finished, the face of the wheels was recessed to represent wheel spokes, then they were fixed onto the axles with high-strength Loctite 638 (Pollymodelengineering.co.uk,2017). This process was determined by combining ideas from a video by Keith Appleton (2017) and model engineer forums, Crossman (2020). As a result of following this method, perfect wheels were created.

Figure 38: Milling the frame sides.

Then the side frames were machined from 4mm hot rolled steel for strength and lower risk of distortion from internal stress. To ensure a matching pair was produced, I riveted the two plates together. The edges were milled in one set-up to maintain parallelism (Figure 38) by cutting in stages, moving the mill head each time, to reach the far ends as detailed by MrCrispin (2014). Next, a vernier height gauge was used to mark out features such as axle box slots ready for milling using consistent datums, which facilitated the critical placement accuracy. Though, the buffer bar mounting holes were not drilled until frame assembly. The aesthetical cut-outs were formed using a large endmill and hand filing to create the correct radii before splitting the finished frames apart.

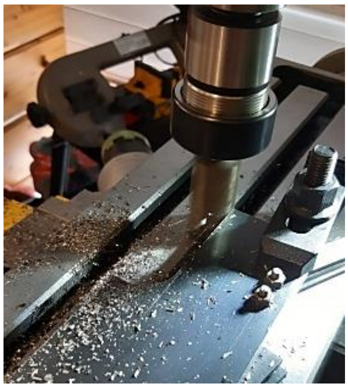

Figure 39: Milling the frame radii.

Once the buffer beams and frame stretchers were completed, the chassis was assembled. This involved marking bracket hole placements on the frames, while the frames were temporarily held together with clamps on a flat granite surface for straightness. This ensured that the frame holes (drilled using a pillar drill for perpendicularity) line up with the bracket holes. This method was based on a video by MrCrispin (2015) and (Evans,1968, page 15).

Figure 40: Setting up the frames ready for drilling the buffer bar bracket holes.

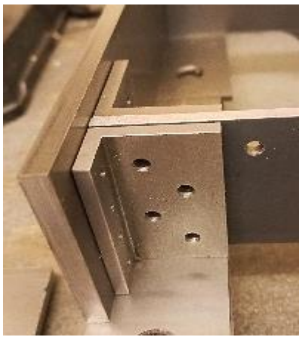

Figure 41: Buffer bar support Brackets.

Figure 42: Drilling the axle box bearing holes.

Another tricky task was making the axle boxes because they required high accuracy to facilitate free running. Six rough blocks were squared up and aligned in the 4-jaw chuck using an accurately Centre drilled point, 2 dead centres and a dial test indicator ready for drilling. This method was attained from Blondihacks (2019). Whilst there are other procedures, this was the most suitable. The drilled hole was reamed for geometrical accuracy and other features such as oil/spring holes were added. The final operation was to mill the retaining slots on either side using an endmill. The depths of these slots were chosen specifically for each axle box to maintain centrality.

Figure 43: milling the crank radii using a rotary table.

Once the axles and bearings were manufactured the cranks were machined. As described in many model engineer forums, the pitch of the axle and crankpin hole must be equal on all cranks to eliminate coupling binding. Acting on this information, I glued all 6 cranks together and drilled the two holes. This ensured equal pitches between holes. Then the retaining grub screw holes were tapped; before I roughed out and machined the radii using files and a rotary table (Figure 43), (Blondihacks,2022).

Figure 44: A finished crank/axle box assembly.

The coupling rods were the hardest components to machine, due to intricate curves and critical crankpin hole spacing, which is equal to the measured distance between the axles on my assembled chassis. These measurements were 150.1 and 150.05mm at the front and rear respectively. Any deviation in this dimension would have caused the rods to bind, so they were drilled on the mill using the dials for accuracy. Then, the connecting joint was milled leaving clearances of ≈ 0.2mm for movement (Clark, 2013), and the oil holes were drilled.

Subsequently the rod width was reduced using the mill, which in turn formed the internal 10mm radii. The external radii were filled rather than milled because my rotary table lacked rigidity which would have resulted in chatter. So, to file to the correct shape, two hardened silver steel “buttons” were used as a guide. This safer method was described by Evans (1968, page 44). Then the respective crankpins and bearings were manufactured ready for assembling the couplings. (Leach, 2006).

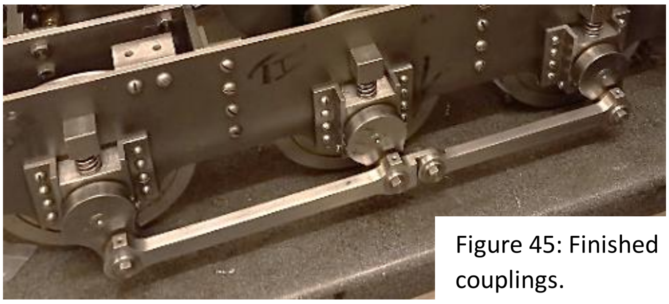
Figure 45: Finished couplings.


Figure 46: Drilling the crankpin holes in the coupling rods.

The motors were connected to the chassis via a lay-shaft double reduction chain drive to reduce the output RPM, as suggested by the engineer that I spoke to. Assembling this system was fraught with difficulty because the chains required the correct tension, which was tricky because chain length is not infinitely variable and all the links must line up. Therefore, I utilized mock-ups and clamps to test the machined frames before the mounting holes were drilled to ensure perfect alignment (Figure 46). Once the motors and chains were setup, I wired up the electronics, adding motor capacitors, fuses and routing the main wiring loom through the chassis to the DN010 controller in the cab. Connectors such as spade and IDC were crimped onto their respective wires and joined as per the wiring diagram, successfully completing the mechanical aspect of the chassis.


Figure 47: Filing the rod radii using “buttons”.

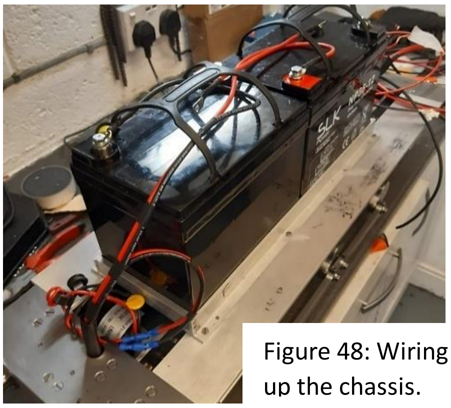
Figure 48: Wiring up the chassis.


Figure 49: Boring the buffer stocks.

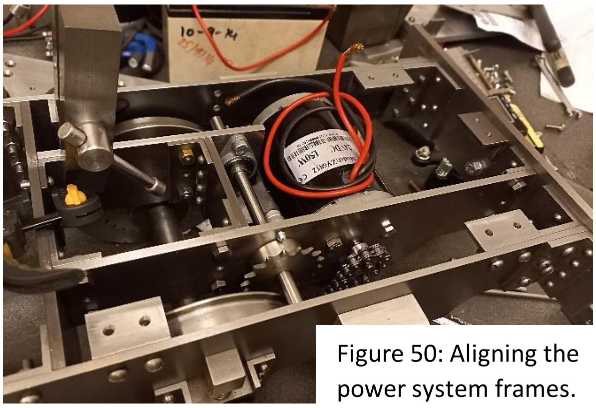
Figure 50: Aligning the power system frames.

With the mechanical systems complete, my attention turned to making aesthetical parts such as buffers. The stocks were mounted centrally in the 4-jaw chuck like the axle boxes but were not removed until all operations were finished to maintain concentricity. The basic shape was turned with a profiling tool and a file was used to form the small front radii to reduce vibrations. Internal features such as the 16mm flat-bottomed hole were created with a boring bar and 16mm endmill for ease. The heads also required a similar hole, along with a tapped hole for a retaining pin, which keeps the head from falling off. To ensure perpendicularity this hole was tapped with a spring Centre and tap in the lathe (IMAT Toole Tech, 2018). The curved face of the head was turned using a rod positioned between the Cross slide and headstock (Figure 51). When the tool was moved toward the centre, the rod increased the distance from the headstock, hence a 150mm radius was formed as illustrated by AlanTurner (2020).


Figure 51: Turning the 150mm radius

The buffers were attached to complete the chassis following a successful manufacturing phase. However, I didn’t meet the deadline specified in my Gantt chart because I remanufactured all inaccurate components to create a high-quality chassis.

Mechanical Performance Testing

Upon completion of the manufacturing stage, I took the chassis to a track to assess the operation of the mechanical systems. I used the table below to evaluate the performance of my locomotive and identify any issues that required rectification.

Testing criteria:	Results:
Traversal of points and curves	The locomotive moved through points and curves with no resistance or derailment due to the integration of bearing clearances and wheel standards, which qualified the accuracy of my drawings (Figure 52)
Max/Min Speed	The chosen 6.6-1 gear ratio reduced the motor speed to an extent. While very low, high torque speeds were attained, the maximum speed attained was greater than expected, which suggests the need for further reduction. However, this would require major redesigns that would only improve performance slightly so the current ratio is sufficient.
Coupling	Through using the correct buffer spacing standards and clevis coupler a secure and reliable connection was achieved with wagons/ coaches.
Power output	The locomotive pulled three people along the track; this surpasses my aim of two people and reinforces the suitability of the gear ratio and 240w motor output power.
Battery capacity	During the 2-hour test run the battery condition monitor never reduced, which suggests that the correct batteries were chosen using Figure 20.
Brakes	The regenerative braking stopped the train over a large distance, so to improve effectiveness I will combine the use of a wagon hand brake.
Suspension	The axle boxes moved freely in the slots, compensating for track irregularities; however, at rest the springs were compressed beyond the expected 3mm so, I have increased the spring stiffness to 30N/mm.
Transmission	The transmission ran freely but two of the chains were slack which caused some vibrations/noise. This issue could not be rectified because moving the layshaft position would create other problems such as misaligned chains or lack of sprocket clearance.

Figure 52: Traversing points.

Figure 53: Traversing a curve.

Overall, the chassis operated well, especially at low speeds where the smoothness of the couplings was observed. This proved that the chosen methods of manufacture were capable of eliminating the risk of binding. Also, the chassis was very rigid, due to the integration of multiple frame stretchers and buffer bar support brackets in my drawings. However, as illustrated by some engineers at the track, the axle box bearing clearance of 0.5mm was too large for a short wheel-base Class 08. I therefore redesigned the bearings to reduce the clearance and increase the flange diameter; which should improve the lifespan of the bearings by reducing radial/axial loads. Through the implementation of these alterations, the final performance of my chassis surpassed many other battery locomotives at the track in relation to power, strength and complexity, which satisfied my mechanical performance objectives.

Aesthetical Performance Testing

To evaluate aesthetical accuracy, I compared my chassis against images because I couldn’t access a model in person. Images of existing models built using commercial drawings, were more valuable than the full-size locomotive for these comparisons as their level of detail is achievable; so I could assess the quality of my researched drawings.

Figure 54: Side profile of the finished chassis.

Figure 55: Side profile of an existing model Keith (2019).

Figure 56: Side profile of the full-size locomotive (wikimedia.org, 2022).

Aesthetically, all of the chassis’ components such as coupling rods look well made because they were manufactured with high tolerances and care, which has resulted in a very sleek finish. However, a clear distinction can be formed between my locomotive and the real Class 08 (Figure 56) in relation to the level of detail and dimensions. For example, the coupling rods are thinner and the wheels are slightly larger than mine. While the aesthetical accuracy of my model is reduced, these dimensions were a compromise for clearance and ease of machining, although there may be some scaling errors from my design process.

Like other 5” gauge models, my chassis is a representation, with streamlined features such as the sandboxes, wheels and horn blocks. Although the replicas in Figure 55 and Figure 10 do incorporate more detail, including access steps, guard irons, leaf springs and brakes. The integration of these characteristics would have improved accuracy, but at the expense of meeting my planned timescale. So, I chose to exclude them because a good-looking locomotive can still be created with fewer details as shown by Figure 8.

Nevertheless, aspects that I integrated, such as the buffers and screw-link coupling replicate their full-scale counterparts to a high degree of precision which contributes to the aesthetics. Ultimately, I created a chassis with the correct overall dimensions and shapes, but without the level of detail to reproduce a Class 08 closely; although this could be improved in the future by painting and adding more features like vacuum hoses retrospectively.

Conclusion

Overall, this project successfully achieved its aim of investigating the production of a 5” gauge locomotive chassis based purely on research, through the utilisation of books, articles, images, and testing. Despite challenges faced during the design and manufacturing process, a smooth-running, powerful chassis was generated that recreated a Class 08 to a satisfactory degree of accuracy.

The process of generating the chassis was difficult because I had to conceive designs and methods of manufacture for every chassis component. This required a great deal of critical resource evaluation and problem solving. Designing each part enabled me to apply my knowledge of maths, while developing my analytical skills. Producing designs for the coupling rods and transmission was especially challenging because many factors had to be incorporated including, the correct shape, dimensions, clearances, tolerances, and machining processes. Inevitably, I made some mistakes with these parts, due to the limitations of my research. These were rectified in subsequent iterations. I also found manufacturing certain components like the buffers and wheels was tricky, because they required intricate radii and precise dimensions which required a lot of thought to determine the order of machining operations, using videos and books for reference.

Overall, the final artefact is very well engineered, as shown by the perfect operation of the mechanical systems on the track. This highlights the suitability of the chosen manufacturing processes and how my machining skills have improved. However, I could have introduced more details and finer scaling into my drawings, which would have improved the accuracy of replication. This lack of detail accentuates flaws in my research, because If I had used the original works drawings for the Class 08, I could have reproduced the exact profiles and scaling. But this would have reduced the scope of my research material. Though, excluding these details ensured that I followed the timescale set out my Gantt chart. This enabled me to focus on each section separately, which in turn improved the quality of the artefact and overall project. Furthermore, this project has provided me with valuable experience in technical drawing, time management, researching and model engineering, all of which will be useful in my future academic and professional pursuits.

In conclusion, the chassis performs exceptionally well and fulfils my objectives. Therefore, my artefact demonstrates the potential for research to produce moderately accurate models. This is significant in the field of model engineering, since this approach creates opportunities for model engineers to develop their knowledge through the development of their own designs for any type of locomotive. Finally, this project has been a demanding but rewarding experience that has enabled me to explore my passion for engineering.

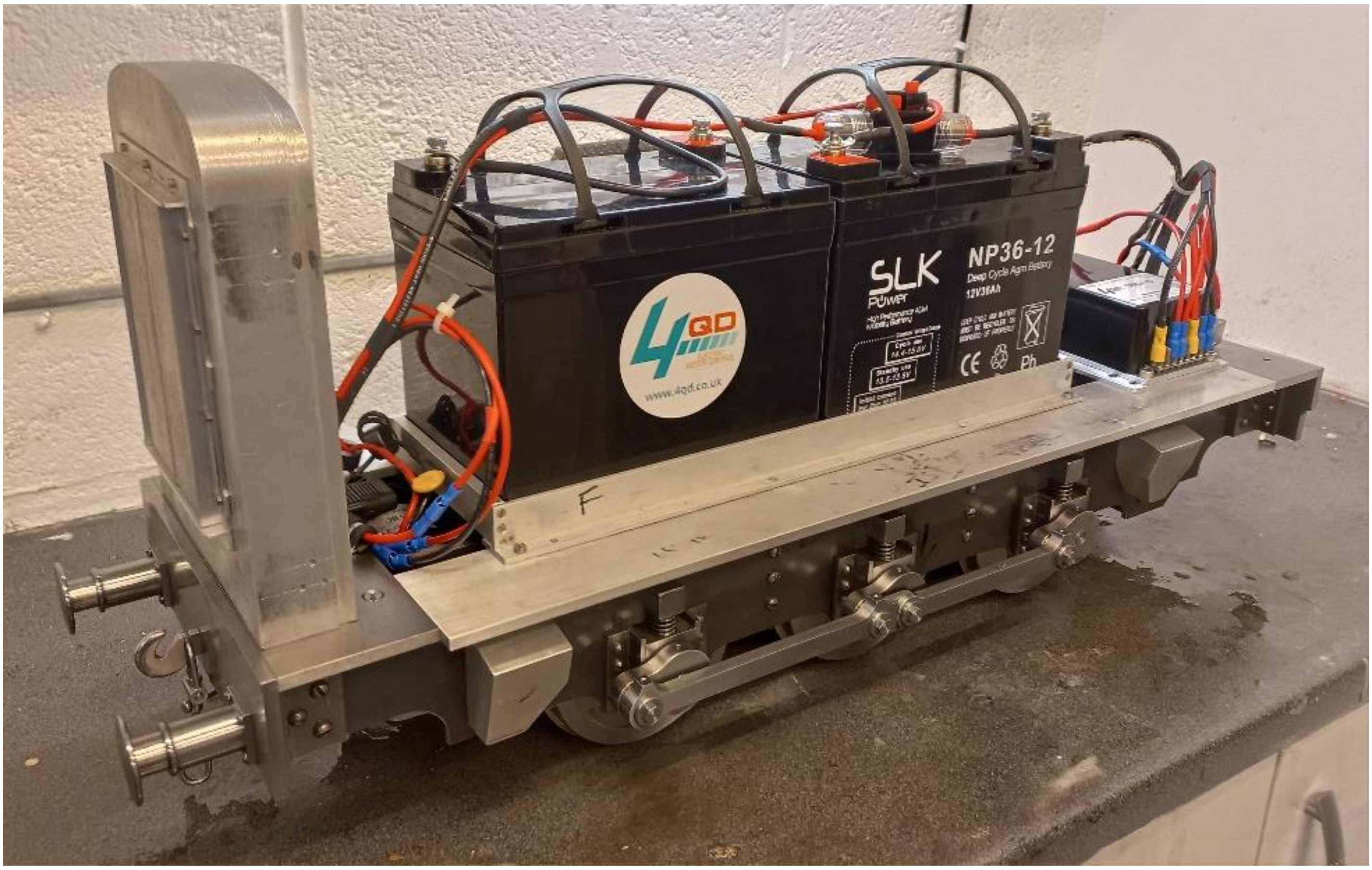

Figure 57 and Figure 58: The finished Class 08 Chassis.

Glossary

Axial load – The direction of a force that acts parallel to the axis of an object.

Axle – A spinning shaft that rotates two or more rotating parts.

Axle Box – The block that attaches the axles to the frames.

Billet – A solid block of metal often round or rectangular in shape.

Buffer bar – The front or rear piece of locomotive frame that the buffers mount to.

Chassis – The section of a locomotive that is below the bodywork and contains most of the mechanical systems.

Chatter – Unsightly marks left by a vibrating lathe or milling cutter.

Concentricity – A measurement of the degree to which two rotating cylindrical objects share the same centre point.

Crank – A mechanical device that consists of an arm attached to a spinning shaft, used to convert rotary motion into a reciprocating motion.

Crankpin – The short round shafts which are fixed onto the crank to drive the coupling rods.

Cross slide – The part of a lathe that moves perpendicular relative to the bed, enabling reduction of the component outer diameter.

Datum – A reference point or line that all dimensions are measured from.

Dead centre – A lathe tool used for supporting the workpiece while in a chuck, or for alignment.

Dial test indicator – A precision measuring tool used to measure small distances or deviations from a desired position.

Endmill – A general milling cutter with cutting edges on the side and face.

Grub Screw – A small threaded fastener with no head.

Headstock – The part of a lathe that supports the main spindle.

Hornblock – The support bracket that increase the frame width to enable positive axle box movement.

Lay shaft – A axle that connects two gears or sprockets together, often to provide secondary speed reduction.

Mandrel – A cylindrical object that is used to hold or support a component during manufacture.

Radial load – A force that acts perpendicular to the axis of rotation.

Rail head – The top surface of a rail on the track.

Reamer – A tool used to enlarge a drilled hole to a precise dimension and geometrical shape.

RPM – The number of times a spinning shaft rotates in one minute.

Sandbox – A container located on the side of a locomotive to provide sand to the rail to increase traction.

Spring Centre – A sprung lathe centre, used for advancing a tap into a drilled hole.

Sprocket – A toothed wheel designed to mesh with a chain.

Stock – The section of a buffer that is mounted to the buffer bar and the head slides into.

Tap – A hand tool that forms a female thread.

Tolerance – A specified deviation from a specified dimension.

Torque – The magnitude of a turning force.

Transmission - A mechanical device used to transmit power from an engine or motor to the wheels of a vehicle or machinery.

Tread profile – The shape of the wheel circumference.

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